

Part III: Interpretation

The virus that is the most related to SARS-CoV-2 (Wuhan) is Bat_CoV_RaTG13, a coronavirus found in bats. It has very high genomic similarity to SARS-CoV-2 (about 96.2% identity), so it clusters very closely with it in phylogenetic analysis. It is therefore a major reference for the study of the evolutionary origin of the virus responsible for the pandemic.

But this type of analysis has several important limits. First, phylogenetic trees are hypotheses and not definitive representations of reality, because they are based on the data used, the assumed evolutionary models. Moreover, differences in sequence length can influence results: if incomplete sequences are included, gaps are introduced that may introduce bias into the alignment. Thus, insertions and deletions reduce the accuracy of multiple sequence alignment when comparing evolutionarily distant viruses. Lastly, the analysis does not always tell us how reliable the relationships shown in the tree are, as it may not include statistical support measures such as bootstrap which limits the interpretation of the robustness of the groupings.